Unit 5 | Chapter 8: Media (Images, Video, Audio & Tables)

Learning Targets:

- Technical: I can select, calculate, and optimize the correct image format (SVG, WebP, JPG, PNG) based on use case and performance. (NV 4.5.1, 4.5.4, 4.5.6)
- Technical: I can implement accessible HTML5 multimedia using <video>, <audio>, and <track> tags, and embed external media using iframes. (NV 4.5.5)
- Technical: I can build semantic data tables using <thead>, <tbody>, <tfoot>, and complex attributes like colspan. (NV 4.4.6)
- Employability: I can demonstrate critical-thinking and problem-solving skills to independently troubleshoot broken media links and layout issues. (WRS 1.2.3)
- Employability: I can demonstrate a positive work ethic, integrity, and diversity awareness by staying on task, legally sourcing media, and ensuring my content is accessible to all users. (WRS 1.1.1, 1.1.2, 1.1.4, 1.1.5)

Compelling Question:

How does a "Digital Architect" balance high-quality sensory media with performance, accessibility, and professional integrity while creating an inclusive web experience?

Score	Performance Indicators
4.0	**I can:** ☐ Design and deploy a high-performance media gallery that uses modern formats (WebP/SVG) and CDN logic to ensure global speed, demonstrating advanced problem-solving. (NV 4.5.4, WRS 1.2.3) ☐ Build a fully accessible, complex data table that uses ARIA labels and semantic headers to pass a Level AA screen-reader audit, demonstrating diversity awareness. (NV 4.6.3, WRS 1.1.5) ☐ Exhibit exceptional workplace readiness by mentoring peers through technical hurdles and ensuring all project assets exhibit strict ethical adherence to copyright laws. (WRS 1.1.2)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	**I can:** ☐ Implement HTML5 video with required attributes for autoplay (muted) and provide subtitles/captions via the <track> tag for diverse users. (NV 4.5.5, WRS 1.1.5) ☐ Calculate dimensions and choose the correct compression tool to reduce file size without quality loss, applying critical thinking to performance. (NV 4.5.6, WRS 1.2.3)

	☐ Calculate and apply colspan and rowspan to create non-linear table structures. ☐ Embed external media (maps, videos) using <iframe> while maintaining professional workplace language and content. (NV 4.5.5, WRS 1.1.4) ☐ Demonstrate *intrinsic motivation* and *punctuality* by completing daily media labs on schedule and maintaining focus without constant redirection. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	**I can** ☐ Define common terminology and identify syntax: (WEB.4.2.1) ☐ SVG, WebP, Lossy vs. Lossless, <iframe>, <video>, <audio>, <track>, <tr>, <td>, <th>, colspan, Compression, Aspect Ratio. ☐ Identify the importance of punctuality and diligence (WRS 1.1.1), probity and ethical adherence (WRS 1.1.2), professional decorum and articulate communication (WRS 1.1.4), inclusive collaboration toward coworkers (WRS 1.1.5), and applying analytical reasoning to resolve issues (WRS 1.2.3).
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.

Unit 10: The Cloud (Version Control & Infrastructure)

Learning Targets:

- Technical: I can select, calculate, and optimize the correct image format (SVG, WebP, JPG, PNG) using Adobe Photoshop and Illustrator based on use case and performance. (NV 4.5.1, 4.5.4, 4.5.6)
- Technical: I can implement accessible HTML5 multimedia using <video>, <audio>, and <track> tags, and embed external media using iframes. (NV 4.5.5)
- Technical: I can build semantic data tables using <thead>, <tbody>, <tfoot>, and complex attributes like colspan. (NV 4.4.6)
- Employability: I can demonstrate analytical reasoning and cognitive flexibility to independently troubleshoot broken media links and layout issues. (WRS 1.2.3)
- Employability: I can demonstrate diligence, probity, and cultural competency by maintaining focus, ethically sourcing media, and ensuring my content is accessible to diverse users. (WRS 1.1.1, 1.1.2, 1.1.4, 1.1.5)

Compelling Question:

How does "The Cloud" move code from a single developer's hard drive to a global audience, and why is version control the "time machine" of modern engineering?

Score	Performance Indicators
4.0	**I can:** ☐ Design and deploy a high-performance media gallery that uses modern formats (WebP/SVG) and CDN logic to ensure global speed, demonstrating advanced analytical reasoning and systematic resolution. (NV 4.5.4, WRS 1.2.3) ☐ Build a fully accessible, complex data table that uses ARIA labels and semantic headers to pass a Level AA screen-reader audit, demonstrating profound cultural competency and empathy for users. (NV 4.6.3, WRS 1.1.5) ☐ Exhibit exceptional workplace readiness by mentoring peers through technical hurdles and ensuring all project assets exhibit strict ethical adherence (probity) to copyright laws. (WRS 1.1.2)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	**I can:** ☐ Implement HTML5 video with required attributes for autoplay (muted) and provide subtitles/captions via the <track> tag to promote an egalitarian user experience. (NV 4.5.5, WRS 1.1.5) ☐ Calculate dimensions and choose the correct Adobe tool to reduce file size

	without quality loss, applying analytical reasoning to performance metrics. (NV 4.5.6, WRS 1.2.3) ☐ Calculate and apply colspan and rowspan to create non-linear table structures. ☐ Embed external media (maps, videos) using <iframe> while demonstrating professional decorum and propriety in content selection. (NV 4.5.5, WRS 1.1.4) ☐ Demonstrate intrinsic motivation and punctuality by completing daily media labs on schedule and maintaining focus without constant redirection. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	**I can** ☐ Define common terminology and identify syntax: (WEB.4.2.1) ☐ Git vs. GitHub, Repository (Repo), Version Control System (VCS), AWS vs. Azure vs. GCP, IaaS/PaaS/SaaS, Scalability vs. Elasticity, Deploy, Merge, Public/Private/Hybrid Cloud, and the "Main" Branch.
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.